# Chapter 7, Problem 4RQ

## Problem

An *RL* circuit has $R = 2\ \Omega$ and $L = 4$ H. The time needed for the inductor current to reach 40 percent of its steady-state value is:

(a) 0.5 s

(b) 1 s

(c) 2 s

(d) 4 s

(e) none of the above

## Step-by-step solution

**Step 1** of 1

We know that

$$I(t) = I_0 e^{\frac{-Rt}{L}}$$

Or, $\frac{Rt}{L} = \ln\left[\frac{I_0}{I(t)}\right]$

Or, $t = \frac{L}{R}\ln\left[\frac{I_0}{I(t)}\right]$

$$= \left(\frac{4F}{2\Omega}\right)\ln\left[\frac{1}{1-0.40}\right] \qquad [40\ \% = 0.40]$$

= 1 S